Level: Bachelor
Program: BE
Course: Microprocessors

POKHARA UNIVERSITY
Semester – Spring

Year: 2020
Full Marks: 70
Pass Marks: 31.5
Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

Section - A: (5×10=50)

Q. N. 1 Design an address decoding circuit for 8085 Microprocessor to interface 2KB ROM, 4KB ROM and 8KB ROM memory modules starting at address 1000H.Mention the requirements for memory interfacing. 10

OR

How can timing diagram in microprocessor support to a software design engineer? Draw a neat and clean timing diagram of OUT 05 H. If one T-state needs 5 nano seconds then compute the total time to execute this instruction in 8085 microprocessor. 10

. N. 2 Explain the different arithmetic and logical instruction of 8085 with its RTL description. Why flags are used in processors? Explain the types of flags in 8085 microprocessor. 10

. N. 3 If the data transfer rate between two peripheral device needs to be improved directly without the involvement of the microprocessor, what would you use? Explain its working mechanism along with necessary diagram. What is the interface standard developed by the Electronic Industries Association in response to the need for the signal and handshake standard between the DTE and DCE? Explain with its features. 10

. N. 4 What are the merits of using segmentation in microprocessors system? Explain. Write an 8085 ALP to count the number of 1 in the given string '101101010' and display the result in C0C0H. 10

N. 5 Write a BSR control word subroutine for 8255 to set bits PC1 and PC57 and reset them after 10ms. Assume control register address= 83H and delay subroutine is available. What are the values of S0 and S1 for Read, Write and Opcode Fetch Operation? 10

Section - B: (1×20=20)

N.6 a) Explain the comparative study among RS 232C and RS 422 A. Why Dual Core and Quad processors are used? Highlights their advantages in microprocessor system. 10

b) Why accumulators are used? Write an assembly language program for 8086 to find the square root of a number. Given that the number is a perfect square and is of two digits. 10

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